

Maxwell's Relation and Thermodynamic Potential

3. Application of Maxwell Relation

First $T - dS$ equation

Let T , and V are independent variable $S = S(T, V)$

$$dS = \left(\frac{\partial S}{\partial T} \right)_V dT + \left(\frac{\partial S}{\partial V} \right)_T dV$$

$$TdS = T \left(\frac{\partial S}{\partial T} \right)_V dT + T \left(\frac{\partial S}{\partial V} \right)_T dV \quad \text{put } \left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial P}{\partial T} \right)_V$$

$$T \cdot dS = C_V dT + T \left(\frac{\partial P}{\partial T} \right)_V dV.$$

Second $T - dS$ Equation

Let T and P are independent variable $S = S(T, P)$.

$$TdS = T\left(\frac{\partial S}{\partial T}\right)_P dT + T\left(\frac{\partial S}{\partial P}\right)_T dP$$

From Maxwell relation

$$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$$

$$TdS = C_P dT - T\left(\frac{\partial V}{\partial T}\right)_P dP$$

Third T-dS Equation:

Let P, V are independent variable $S = S(P, V)$

$$dS = \left(\frac{\partial S}{\partial P}\right)_V dP + \left(\frac{\partial S}{\partial V}\right)_P dV$$

$$TdS = T\left(\frac{\partial S}{\partial P}\right)_V dP + T\left(\frac{\partial S}{\partial V}\right)_P dV$$

$$= T\left(\frac{\partial S}{\partial T}\right)_V \left(\frac{\partial T}{\partial P}\right)_V dP + T\left(\frac{\partial S}{\partial T}\right)_P \left(\frac{\partial T}{\partial V}\right)_P dV$$

$$= C_V \left(\frac{\partial T}{\partial P}\right)_V dP + C_P \left(\frac{\partial T}{\partial V}\right)_P dV$$

The First Energy Equation

Let T and V are independent variable $U = U(T, V)$

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

From first law of thermodynamics.

$$dU = TdS - PdV$$

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial S}{\partial V}\right)_T - P$$

Using Maxwell relation $T\left(\frac{\partial S}{\partial V}\right)_T = T\left(\frac{\partial P}{\partial T}\right)_V$

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial P}{\partial T}\right)_V - P$$

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

$$= C_v dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$

$$\left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial P}{\partial T} \right)_V - P$$

$$dU = C_v dT + \left(T \left(\frac{\partial P}{\partial T} \right)_V - P \right) dV$$

Second Energy Equation

$$dU = TdS - PdV$$

$$\left(\frac{\partial U}{\partial P} \right)_T = T \left(\frac{\partial S}{\partial P} \right)_T - P \left(\frac{\partial V}{\partial P} \right)_T$$

Using Maxwell relation

$$\left(\frac{\partial S}{\partial P} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_P$$

$$\left(\frac{\partial U}{\partial P} \right)_T = -T \left(\frac{\partial V}{\partial T} \right)_P - P \left(\frac{\partial V}{\partial P} \right)_T$$

This is popularly known as second energy equation

Application of second energy equation

If U is function of independent variable of T and P .

$$U = U(T, P)$$

$$dU = \left(\frac{\partial U}{\partial T} \right)_P dT + \left(\frac{\partial U}{\partial P} \right)_T dP$$

$$dU = \left(\frac{\partial U}{\partial T} \right)_P dT - \left[T \left(\frac{\partial V}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial P} \right)_T \right] dP$$

Example: From relation $dU = TdS - PdV$

Derive Maxwell relation $\left(\frac{\partial T}{\partial V} \right)_S = - \left(\frac{\partial P}{\partial S} \right)_V$

Solution: $dU = TdS - PdV$ $\left(\frac{\partial U}{\partial V} \right)_S = -P$, $\left(\frac{\partial U}{\partial S} \right)_V = T$

Hence U is exact differential

$$\left[\frac{\partial}{\partial S} \left(\frac{\partial U}{\partial V} \right)_S \right]_V = \left[\frac{\partial}{\partial V} \left(\frac{\partial U}{\partial S} \right)_V \right]_S - \left(\frac{\partial P}{\partial S} \right)_V = \left(\frac{\partial T}{\partial V} \right)_S$$

Example: A real gas which obey van der Waal's equation of state are kept in container which has temperature T_0 and volume V_0 . if volume of container changes to V such that temperature of gas become T what is change in entropy?

Solution: Assume C_V is specific heat of constant volume

For van der Waal's gas

$$\left(P + \frac{a}{V^2} \right) (V - b) = RT$$

From first $T - dS$ equation

$$TdS = C_V dT + \left(\frac{\partial P}{\partial T} \right)_V dV$$

$$\left(\frac{\partial P}{\partial T} \right)_V = \frac{R}{V - b}$$

$$dS = C_V \int_{T_0}^T \frac{dT}{T} + \int_{V_0}^V \frac{R}{(V - b)} dV \Rightarrow S = C_V \ln \left(\frac{T}{T_0} \right) + R \ln \left(\frac{V - b}{V_0 - b} \right) + S_0$$

where S_0 is integration constant

Example: For Vander wall gases, prove that

$$\left(\frac{\partial U}{\partial V} \right)_T = \frac{a}{V^2} \text{ where } U \text{ is internal energy.}$$

Solution: From first energy equation

$$\left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial P}{\partial T} \right)_V - P \quad \text{-----(i)}$$

$$\left(P + \frac{a}{V^2} \right) (V - b) = RT$$

$$P = \frac{RT}{V - b} - \frac{a}{V^2}$$

$$\left(\frac{\partial P}{\partial T} \right)_V = \frac{R}{(V - b)} \text{ put the value of } \left(\frac{\partial P}{\partial T} \right)_V \text{ in equation (i)}$$

$$\left(\frac{\partial U}{\partial V} \right)_T = \frac{a}{V^2}$$

Example: Prove that

$$(a) \left(\frac{\partial C_V}{\partial V} \right)_T = T \left(\frac{\partial^2 P}{\partial T^2} \right)_V$$

$$(b) \left(\frac{\partial C_P}{\partial P} \right)_T = -T \left(\frac{\partial^2 V}{\partial T^2} \right)_P$$

Solution: (a) we know $C_V = T \left(\frac{\partial S}{\partial T} \right)_V$

$$\left(\frac{\partial C_V}{\partial V} \right)_T = T \left[\frac{\partial}{\partial V} \left(\frac{\partial S}{\partial T} \right)_V \right]_T = T \left[\frac{\partial}{\partial T} \left(\frac{\partial S}{\partial V} \right)_T \right]_V$$

using Maxwell relation

$$\left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial P}{\partial T} \right)_V \text{ one can get } T \left[\frac{\partial}{\partial T} \left(\frac{\partial S}{\partial V} \right)_T \right]_V = T \left[\frac{\partial}{\partial T} \left(\frac{\partial P}{\partial T} \right)_V \right]_V, \left(\frac{\partial C_V}{\partial V} \right)_T = T \left(\frac{\partial^2 P}{\partial T^2} \right)_V$$

$$(b) C_P = T \left(\frac{\partial S}{\partial T} \right)_P, \left(\frac{\partial C_P}{\partial P} \right)_T = T \left[\frac{\partial}{\partial P} \left(\frac{\partial S}{\partial T} \right)_P \right]_T = T \left[\frac{\partial}{\partial T} \left(\frac{\partial S}{\partial P} \right)_T \right]_P$$

Use Maxwell relation

$$\left(\frac{\partial S}{\partial P} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_P, \left(\frac{\partial C_P}{\partial P} \right)_T = -T \left(\frac{\partial^2 V}{\partial T^2} \right)_P$$

Example: If α_p is thermal expansivity at constant pressure and K_T isothermal compressibility then prove that

$$(i) \left(\frac{\partial S}{\partial P} \right)_T = -V\alpha_p \quad (ii) \left(\frac{\partial P}{\partial T} \right)_V = \frac{\alpha_p}{K_T} \quad (iii) C_P - C_V = \frac{TV\alpha_p^2 P}{K_T}$$

Solution: (i) From Maxwell relation

$$\left(\frac{\partial S}{\partial P} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_P$$

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P$$

$$\left(\frac{\partial S}{\partial P} \right)_T = -V \left[\frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P \right], \left(\frac{\partial S}{\partial P} \right)_T = -\alpha V$$

$$(ii) \left(\frac{\partial P}{\partial T} \right)_V = \left(\frac{\partial S}{\partial V} \right)_T$$

$$\left(\frac{\partial P}{\partial T}\right)_V \left(\frac{\partial V}{\partial P}\right)_T \left(\frac{\partial T}{\partial V}\right)_P = -1$$

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{-1}{\left(\frac{\partial V}{\partial P}\right)_T \left(\frac{\partial T}{\partial V}\right)_P} = \frac{-\frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_P}{\frac{1}{V} \left(\frac{\partial V}{\partial P}\right)_T}, \quad \left(\frac{\partial P}{\partial T}\right)_V = \frac{-\alpha}{K_T}$$

$$dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV \quad dS = \frac{C_V}{T} dT + \left(\frac{\partial S}{\partial V}\right)_T dV$$

$$\left(\frac{\partial S}{\partial T}\right)_P = \frac{C_V}{T} + \left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P \quad \frac{C_P}{T} = \frac{C_V}{T} + V \alpha_P \left(\frac{\partial S}{\partial V}\right)_T$$

Use Maxwell relation

$$\frac{C_P}{T} = \frac{C_V}{T} + V \alpha_P \left(\frac{\partial P}{\partial T}\right)_V \quad \frac{C_P}{T} = \frac{C_V}{T} + V \alpha_P \left(\frac{\alpha_P}{K_T}\right)$$

$$C_P - C_V = \frac{TV \alpha_P^2}{K_T}$$

Example: Prove that

(a) $C_P - C_V = T \left(\frac{\partial P}{\partial T}\right)_V \left(\frac{\partial V}{\partial T}\right)_P$

(b) For the Vander Waal's gas prove that

$$C_P - C_V = R \left(1 + \frac{2a}{RTV}\right)$$

Solution: (a) $C_P - C_V = \left(\frac{\partial Q}{\partial T}\right)_P - \left(\frac{\partial Q}{\partial T}\right)_V = T \left(\frac{\partial S}{\partial T}\right)_P - T \left(\frac{\partial S}{\partial T}\right)_V \quad \text{---- (A)}$

$$S = S(T, V)$$

$$dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV, \quad \left(\frac{\partial S}{\partial T}\right)_P = \left(\frac{\partial S}{\partial T}\right)_V + \left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P$$

Put the value of $\left(\frac{\partial S}{\partial T}\right)_P$ in equation (A)

$$C_P - C_V = T \left(\frac{\partial S}{\partial T}\right)_V + T \left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P - T \left(\frac{\partial S}{\partial T}\right)_V \Rightarrow C_P - C_V = T \left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P$$

$$= T \left(\frac{\partial P}{\partial T} \right)_V \left(\frac{\partial V}{\partial T} \right)_P \because \left[\left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial P}{\partial T} \right)_V \right]$$

(b) For van der Waal's gas

$$\left(P + \frac{a}{V^2} \right) (V - b) = RT \quad \text{---- (B)}$$

$$\left(P + \frac{a}{V^2} \right) = \frac{RT}{(V - b)} \rightarrow \text{differentiate w.r.t. to } T$$

$$\left(\frac{\partial P}{\partial T} \right)_V = \frac{R}{V - b}$$

Differentiate (B) with respect to V

$$-\frac{2a}{V^3} \left(\frac{\partial V}{\partial T} \right)_P = -\frac{RT}{(V - b)^2} \left(\frac{\partial V}{\partial T} \right)_P + \frac{R}{V - b}$$

$$\left(\frac{\partial V}{\partial T} \right)_P = \frac{\frac{R}{V - b}}{\frac{RT}{(V - b)^2} - \frac{2a}{V^3}}$$

Substituting the value $\left(\frac{\partial P}{\partial T} \right)_V \left(\frac{\partial V}{\partial T} \right)_P$ in equation

$$C_P - C_V = T \left(\frac{\partial S}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P \quad C_P - C_V = T \frac{\frac{R}{(V - b)} \left(\frac{R}{V - b} \right)}{\frac{RT}{(V - b)^2} - \frac{2a}{V^3}} = \frac{R}{1 - \left(\frac{2a}{V^3} \right) \frac{(V - b)^2}{RT}}$$

$$C_P - C_V = \frac{R}{1 - \left(\frac{2a}{V^3} \right) \frac{V^2}{RT}} = R \left(1 - \frac{2a}{RTV} \right)^{-1}, \quad C_P - C_V = R \left(1 + \frac{2a}{RTV} \right)$$

Example: From $C_P - C_V = T \left(\frac{\partial P}{\partial T} \right)_V \left(\frac{\partial V}{\partial T} \right)_P$

Prove $C_P - C_V = TE\alpha^2 V$ where E is bulk modulus of elasticity and α is coefficient of volume expansion.

Solution: Let $C_P - C_V = T \left(\frac{\partial P}{\partial T} \right)_V \left(\frac{\partial V}{\partial T} \right)_P$

$$P = P(T, V)$$

$$dP = \left(\frac{\partial P}{\partial T} \right)_V dT + \left(\frac{\partial P}{\partial V} \right)_T dV$$

For constant pressure $dP = 0$

$$\left(\frac{\partial P}{\partial T} \right)_V dT = - \left(\frac{\partial P}{\partial V} \right)_T dV$$

$$\left(\frac{\partial P}{\partial T} \right)_V = - \left(\frac{\partial P}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P$$

$$C_P - C_V = -T \left(\frac{\partial P}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P^2$$

$$E = -V \left(\frac{\partial P}{\partial V} \right)_T \text{ and } \alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P$$

$$C_P - C_V = -T \frac{V}{V} \left(\frac{\partial P}{\partial V} \right)_T \frac{V^2}{V} \left(\frac{\partial V}{\partial T} \right)_P^2 = -T \left(\frac{E}{V} \right) V^2 \alpha^2$$

$$C_P - C_V = -TVE\alpha^2$$

Example: If Helmholtz free energy for radiation is given by

$$F = -\frac{8\pi^5 K_B^4 T^4}{45h^3 c^3} V$$

(a) What is radiation pressure?

(b) If S entropy of system prove that specific heat at constant volume is given by $C_V = 3S$

Solution: (a) $dF = -SdT - PdV$

$$P = - \left(\frac{\partial F}{\partial V} \right)_T = \frac{8\pi^5 K_B^4 T^4}{45h^3 c^3}$$

$$(b) S = - \left(\frac{\partial F}{\partial T} \right)_V = \frac{32\pi^5 K_B^4}{45h^3 c^3} VT^3$$

$$C_V = T \left(\frac{\partial S}{\partial T} \right)_V = T \cdot 3 \cdot \left(\frac{32\pi^5 K_B^4}{45h^3 c^3} \right) VT^2 = 3 \cdot \left(\frac{32\pi^5 K_B^4}{45h^3 c^3} \right) VT^3$$

$$C_V = 3S$$

Example: The internal energy E of a system is given by $E = \frac{bS^3}{VN}$ where b is constant and other

symbols have their used meaning.

(a) Find the temperature of system

(b) Find Pressure of system

Solution: From first law of thermodynamics

$$TdS = dU + PdV \quad U = E = \frac{bS^3}{VN} \quad dU = TdS - PdV$$

$$(a) \quad T = \left(\frac{\partial U}{\partial S} \right)_V \quad T = \frac{3bS^2}{VN}$$

$$(b) \quad P = - \left(\frac{\partial U}{\partial V} \right)_S \quad P = - \left(- \frac{bS^3}{V^2 N} \right) \quad P = \frac{bS^3}{V^2 N}$$

Example: Consider an Ideal gas where entropy is given by

$$S = \frac{n}{2} \left[\sigma + 5R \ln \frac{U}{n} + 2R \ln \frac{V}{n} \right]$$

where n = number of moles, R = universal gas constant, U = internal energy

V = volume and σ = constant

(a) Calculate specific heat at constant pressure and volume

(b) Prove that internal energy is given by $U = \frac{5}{2} PV$

Solution: (a) From first law of thermodynamics

$$TdS = dU - PdV, \quad dS = \frac{1}{T} dU - \frac{P}{T} dV \quad \left(\frac{\partial S}{\partial U} \right)_V = \frac{1}{T}$$

$$\frac{n}{2} \frac{5R}{U} = \frac{1}{T} \quad U = \frac{5}{2} nRT$$

$$C_V = \left(\frac{\partial U}{\partial T} \right) = \frac{5}{2} nR \Rightarrow C_P = C_V + R \Rightarrow C_P = \frac{7}{2} nR$$

$$(b) \quad U = \frac{5}{2} nRT \Rightarrow \frac{U}{V} = \frac{5}{2} \frac{nRT}{V}$$

$$PV = nRT \Rightarrow V = \frac{nRT}{P}$$

$$\frac{U}{V} = \frac{5}{2} P \Rightarrow U = \frac{5}{2} PV$$

Example: Using the equation of state $PV = nRT$ and the specific heat per mole $C_v = \frac{3R}{2}$ for

monatomic ideal gas

(a) Find Entropy of given system.

(b) Find free energy of given system

Solution: (a) $dU = nC_v dT$, $p = \frac{nRT}{V}$ $C_v = \frac{3R}{2}$

$$TdS = dU + PdV, \quad dS = \frac{dU}{T} + \frac{p}{T}dV \quad \text{or} \quad ds = nC_v \frac{dT}{T} + nR \frac{dV}{V}$$

$$S = \frac{3}{2} NR \ln T + N \ln V + S_0 \text{ where } S_0 \text{ is constant}$$

(b) $F = U - TS$

$$= \frac{3nRT}{2} - \left(\frac{3nRT}{2} \ln T - nRT \ln V \right) + F_0 \text{ where } F_0 = T \cdot S_0 \text{ is again constant}$$

Example: From electromagnetic theory Maxwell found that the pressure P from an isotropic radiation equal to $\frac{1}{3}$ the energy density i.e. $P = \frac{1}{3} \frac{U}{V}$ where V is volume of the cavity using the

first energy equation prove that

Energy density u is proportional to T^4 .

Solution: $\left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial S}{\partial V} \right)_T - P = T \left(\frac{\partial P}{\partial T} \right)_T - P$

$$P = \frac{1}{3} u \quad u = \frac{U}{V}$$

$$= \left(\frac{\partial P}{\partial T} \right)_V = \frac{1}{3} \left(\frac{\partial U}{\partial T} \right)_V \quad \left(\frac{\partial U}{\partial V} \right)_T = \frac{T}{3V} \left(\frac{\partial U}{\partial T} \right)_V - \frac{U}{3V}$$

$$\left(\frac{\partial U}{\partial V} \right)_T = u \quad u = \frac{T}{3} \left(\frac{du}{dT} \right)_V - \frac{1}{3} u$$

$$T \frac{du}{dT} = 4u = \frac{du}{u} = 4 \frac{dT}{T}$$

$$u \propto T^4 = u = \alpha T^4 \quad \text{where } \alpha \text{ is a constant.}$$